

Learning AI Early: A Meta-Synthesis of Literature Reviews on AI Education at the Elementary Level

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Introduction

- Increased need for AI education in K-12 settings as a response to the widespread impact of AI in society
- Focus on AI literacy – knowledge and critical awareness needed to make sense of AI technologies
- Researchers often treat K-12 students as a homogenous group despite differences
- This meta-synthesis leverages existing literature to identify patterns, gaps, and tensions in how AI is taught to elementary students

Research Questions

RQ.
1

How is AI literacy defined and operationalized in elementary education?

RQ.
2

In what ways do these programs differ from traditional computing education initiatives?

RQ.
3

What adjustments have been made to adapt AI literacy content for younger audiences?

Definitions

(CSTA & AI4K12, 2025)

- **Artificial Intelligence (AI):** a system that uses training data to produce outputs like predictions, recommendations, or decisions that are considered similar to actions that reflect “intelligence” in humans.
- **Generative AI:** technologies that can generate new text, images, video, and/or audio in response to a prompt,
- **Machine Learning (ML):** a subfield of AI that combines learning algorithms with training data to create tools that can be used to complete tasks.

Databases and Search Terms

- Web of Science, EBSCO, and Scopus
- Keywords:
 - AI Literacy and AI education: ("AI literacy" OR "artificial intelligence literacy" OR "AI education" OR "artificial intelligence education")
 - Educational: ("elementary" OR "K–12" OR "primary")
 - Review: ("systematic review" OR "scoping review" OR "literature review")

Identifying Relevant Reviews

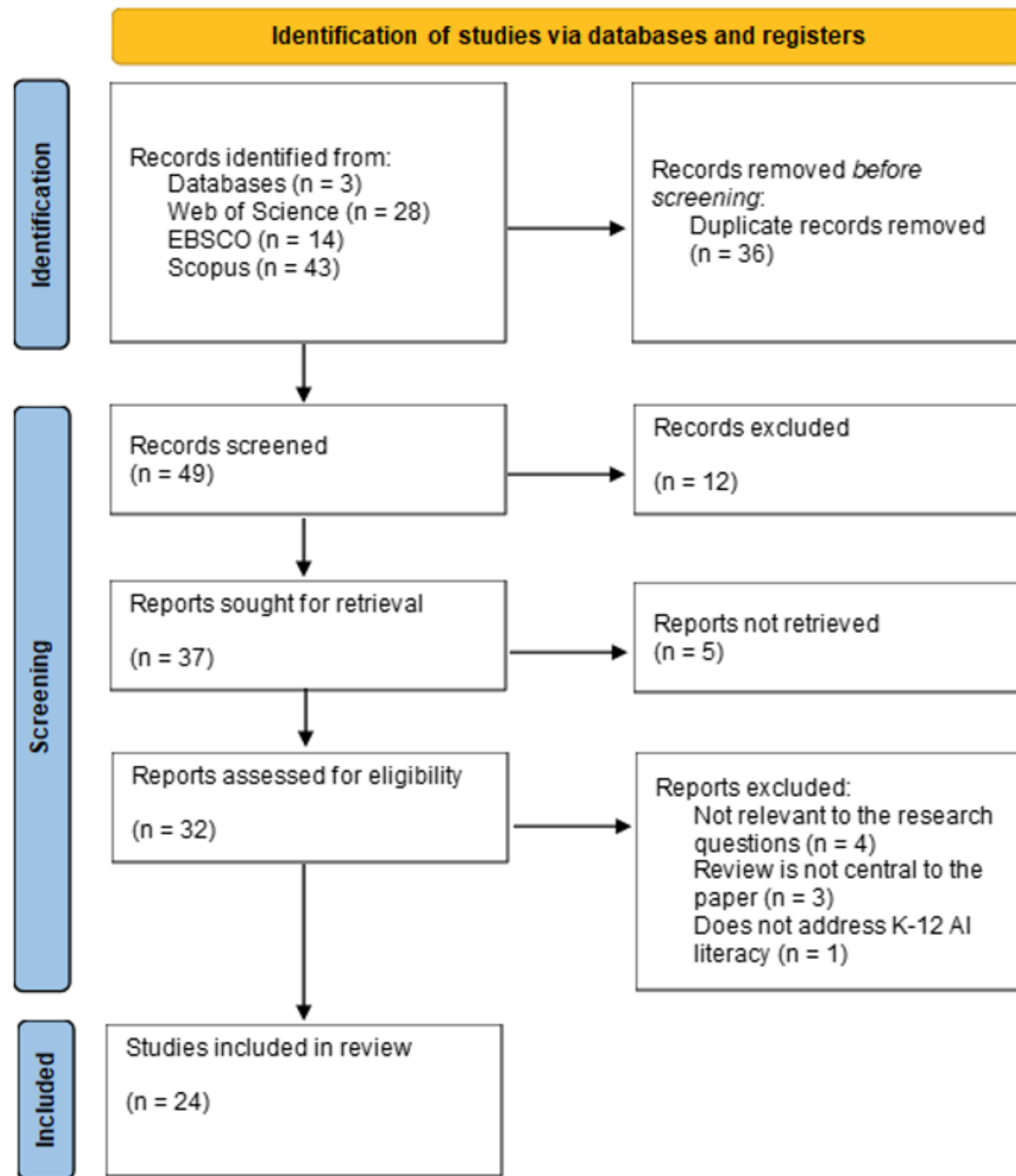
Inclusion Criteria

- Peer-reviewed journal or conference papers
- Published between 2018 and Dec 2025
- Written in English
- Available in full text

Exclusion Criteria

- Not related to K-12 AI Education
- Review was not the focus of the article
- Does not address AI literacy
- Findings not relevant to the RQs

Search Protocol



How AI Literacy is Defined (RQ₁)

- No clear definition of AI literacy or its competencies
- Frameworks often group K-12, no frameworks exclusively for elementary education
 - Ex: Yim (2024) found only four AI literacy frameworks for K-12 learners
- AI literacy is defined as a multidimensional construct that integrates [1-3]:
 - Conceptual understanding (e.g., data, models, learning, and representation)
 - Practical skills
 - Ethical awareness

How AI Literacy is Operationalized (RQ1)

- AI literacy is operationalized through playful, project-based, and culturally responsive learning activities [1-2]
 - Help young children understand, interrogate, and responsibly use AI systems
 - Often embedded within familiar school subjects
 - E.g., model-building, structured experimentation, embodied simulations, or guided
- Informal learning environments play a significant role in shaping young children's early AI understandings [2-4]

How AI Programs Differ from Traditional Computing Education Initiatives (RQ2)

- Teachers generally lack computer science knowledge to teach in-depth coding concepts [1]
- Young learners often struggle with coding-intensive AI learning activities and lack the intellectual ability to benefit from them [2-3]
- Reviews recommend designing flexible and accessible AI learning experiences [4-5]
- AI education efforts encourage data-driven approaches as apposed to traditional computing education [5]

Adjustments to Adapt AI Literacy Content for Younger Audiences (RQ₃)

- Simplifying AI concepts at the cost of conceptual depth or “rigor”
 - Ex: students interacting with a simplified AI tool to teach AI algorithm functionality [1]
 - Transparency at the cost of mathematical rigor
- Age-appropriate pedagogical approaches to AI education for young children [2-3]:
 - Storytelling, role-playing, embodied learning, etc.
 - Introduce foundational ideas such as perception, classification, pattern recognition, and model behavior

Gaps and Future Work

- Gaps in the literature:
 - Lack of frameworks tailored to elementary learners
 - Conceptual rigor is often sacrificed for accessibility
 - Research frequently groups K-12 learners
- Future work:
 - Designing child-centered AI literacy frameworks
 - Strengthening teacher preparation
 - More research on developing AI literacy for young learners

Limitations

- Findings depend on how reviews selected and interpreted literature
- Some newer or less visible studies may not have been included
- More firsthand and systematic research of AI literacy development for young children is needed

Conclusion

By emphasizing the unique needs of elementary students, this meta-synthesis highlights the importance of building AI education that is cognitively appropriate, ethically informed, and capable of supporting children's long-term growth as critical and responsible citizens in an AI-enabled world.

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